Group Topic: *Predicting Student Dropout Risk Using Academic and Behavioral Data* **Lab No**.

Git Repo/ Colab Link: https://github.com/dgbb-prog/student-dropout-bigdata-datacleaning **Date:** *04/24/26*

Objectives

The purpose of this lab is to prepare the OULAD (Open University Learning Analytics Dataset) for predictive modeling by performing data cleaning, preprocessing, and feature engineering. This activity aims to understand student dropout patterns through academic and behavioral data, and to create visualizations that reveal key insights about dropout risk factors.

The specific focus of the Data Loading and Initial Inspection role was to establish the development environment, successfully ingest the 10.9-million-row dataset, and perform a preliminary audit of the raw data to ensure its integrity and volume met the project's requirements.

Tasks Completed:

The following tasks were accomplished during this week as part of data cleaning and preprocessing:

Task 1 — Install and Import Libraries

We initialized the environment by importing `pandas` for data manipulation, `matplotlib` and `seaborn` for visualization, and `zipfile` for file management.

```python
# Install required libraries
!pip install kaggle --quiet

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
import os
import zipfile

warnings.filterwarnings('ignore')

plt.rcParams['figure.figsize'] = (12, 6)
plt.rcParams['font.size'] = 12
sns.set_style('whitegrid')
sns.set_palette('Set2')

print('All libraries imported successfully!')
```

```
All libraries imported successfully!
```

Task 2 — Upload the Dataset

The OULAD dataset was sourced from the official Open University repository. The dataset was provided as a ZIP archive containing 7 individual CSV files.

```
from google.colab import files

print('Please upload the OULAD zip file...')
print('Download it first from: https://analyse.kmi.open.ac.uk/open-dataset')
uploaded = files.upload()
```

Task 3 — Extract the ZIP File

The uploaded archive was extracted into a dedicated directory (`/content/oulad/`). We verified the extraction by listing the files and their respective sizes in MB.

```
zip_filename = list(uploaded.keys())[0]

with zipfile.ZipFile(zip_filename, 'r') as z:
    z.extractall('/content/oulad/')

print('Extracted files:')
for f in os.listdir('/content/oulad/'):
    size = os.path.getsize(f'/content/oulad/{f}') / 1024 / 1024
    print(f'  {f}  ({size:.2f} MB)')
```

Task 4 — Load All 7 CSV Files

Each CSV file was read into a pandas DataFrame. We verified the ingestion by printing the row and column counts for each table, confirming the dataset meets the project size requirements.

```
base_path = '/content/oulad/'

student_info       = pd.read_csv(base_path + 'studentInfo.csv')
student_vle        = pd.read_csv(base_path + 'studentVle.csv')
student_assessment = pd.read_csv(base_path + 'studentAssessment.csv')
student_reg        = pd.read_csv(base_path + 'studentRegistration.csv')
assessments        = pd.read_csv(base_path + 'assessments.csv')
vle                = pd.read_csv(base_path + 'vle.csv')
courses            = pd.read_csv(base_path + 'courses.csv')

print('=== DATASET LOADED SUCCESSFULLY ===')
print(f'studentInfo       : {student_info.shape[0]:,} rows x {student_info.shape[1]} columns')
print(f'studentVle        : {student_vle.shape[0]:,} rows x {student_vle.shape[1]} columns')
print(f'studentAssessment : {student_assessment.shape[0]:,} rows x {student_assessment.shape[1]} columns')
print(f'studentReg        : {student_reg.shape[0]:,} rows x {student_reg.shape[1]} columns')
print(f'assessments       : {assessments.shape[0]:,} rows x {assessments.shape[1]} columns')
print(f'vle               : {vle.shape[0]:,} rows x {vle.shape[1]} columns')
print(f'courses           : {courses.shape[0]:,} rows x {courses.shape[1]} columns')

total_rows = (student_info.shape[0] + student_vle.shape[0] +
              student_assessment.shape[0] + student_reg.shape[0] +
              assessments.shape[0] + vle.shape[0] + courses.shape[0])
print(f'\nTOTAL ROWS (all files combined): {total_rows:,}')
```

```python
print('=== MISSING VALUES PER FILE ===')

datasets = {
    'studentInfo': student_info,
    'studentVle': student_vle,
    'studentAssessment': student_assessment,
    'studentRegistration': student_reg
}

for name, df in datasets.items():
    missing = df.isnull().sum()
    missing = missing[missing > 0]
    if len(missing) > 0:
        print(f'\n{name}:')
        print(missing)
    else:
        print(f'\n{name}: No missing values found.')
```

```
=== MISSING VALUES PER FILE ===

studentInfo:
imd_band     1111
dtype: int64

studentVle: No missing values found.

studentAssessment:
score     173
dtype: int64

studentRegistration:
date_registration          45
date_unregistration     22521
dtype: int64
```

Task 5 — Preview Main Dataset

We focused on `studentInfo.csv` to inspect the demographics and the distribution of our target variable, `final_result`.

```
=== DATASET LOADED SUCCESSFULLY ===
studentInfo       : 32,593 rows x 12 columns
studentVle        : 10,655,280 rows x 6 columns
studentAssessment : 173,912 rows x 5 columns
studentReg        : 32,593 rows x 5 columns
assessments       : 206 rows x 6 columns
vle               : 6,364 rows x 6 columns
courses           : 22 rows x 3 columns

TOTAL ROWS (all files combined): 10,900,970
```

```python
print('=== studentInfo.csv Preview ===')
display(student_info.head())
print()
print('Columns:', list(student_info.columns))
print()
print('Target variable (final_result) distribution:')
print(student_info['final_result'].value_counts())
```

=== studentInfo.csv Preview ===

	code_module	code_presentation	id_student	gender	region	highest_education	imd_band	age_band	num_of_prev_attempts	studied_credits	disability	final_result
0	AAA	2013J	11391	M	East Anglian Region	HE Qualification	90-100%	55<=	0	240	N	Pass
1	AAA	2013J	28400	F	Scotland	HE Qualification	20-30%	35-55	0	60	N	Pass
2	AAA	2013J	30268	F	North Western Region	A Level or Equivalent	30-40%	35-55	0	60	Y	Withdrawn
3	AAA	2013J	31604	F	South East Region	A Level or Equivalent	50-60%	35-55	0	60	N	Pass
4	AAA	2013J	32885	F	West Midlands Region	Lower Than A Level	50-60%	0-35	0	60	N	Pass

```
Columns: ['code_module', 'code_presentation', 'id_student', 'gender', 'region', 'highest_education', 'imd_band', 'age_band', 'num_of_prev_attempts', 'studied_credits', 'disability', 'final_result'

Target variable (final_result) distribution:
final_result
Pass           12361
Withdrawn      10156
Fail            7052
Distinction     3024
Name: count, dtype: int64
```

Results and Analysis

The loading phase confirmed a massive dataset totaling *10,900,970 rows*, providing a robust foundation for large-scale statistical modeling. A key finding from the initial inspection of the `studentInfo` table revealed that *"Withdrawn"* students *(10,156)* account for approximately 31% of the demographic data, ensuring the model has enough data to learn dropout patterns. Furthermore, the missing value audit identified 1,111 nulls in the `imd_band` column and significant gaps in `date_unregistration`, which informs the next steps for data imputation.

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```
=== studentInfo.csv Preview ===
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```
Columns: ['code_module', 'code_presentation', 'id_student', 'gender', 'region', 'highest_education', 'imd_band', 'age_band', 'num_of_prev_attempts', 'studied_credits', 'disability', 'final_result']

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final_result
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Withdrawn     10156
Fail           7052
Distinction    3024
Name: count, dtype: int64
```

Challenges and Solutions

- **Challenge 1:** Loading the `studentVle.csv` file caused significant performance lag and high RAM usage due to its size of over 10 million records.
- **Solution 1:** The file was loaded using standard pandas functions without immediate heavy transformations, deferring intensive computation to the dedicated cleaning and merging phases.
- **Challenge 2:** The relational links between the seven different CSV files were initially complex to navigate.
- **Solution 2:** By referencing the official OULAD documentation, the primary and foreign keys were mapped to clarify how the tables interact for future merging.

Conclusion

The objectives for this initial phase were successfully achieved, resulting in a fully operational data environment. The process ensured that nearly 11 million rows of data (approximately 10.9 million) were correctly ingested, meeting the necessary scale for advanced analytics. Key insights regarding the target variable distribution and identified data gaps have been documented and shared with the team, providing a reliable roadmap for the cleaning and preprocessing tasks.